Frontline Controller In-field Setup

Field Process Guide

Summary

The Frontline Mark I Controller has a default configuration from the factory that allows it to start up and interact with field installation and support personnel. However, beyond this limited factory functionality, for it to successfully connect to the Frontline cloud service and to support activation and hydration of the premises it is installed at, it must be successfully configured and set up.

This document summarizes the field process to use to run setup on a controller, whether for the first time or otherwise.

Overview

At the time of this writing and for firmware release `3.0.0` and later, setup is handled interactively over a text-based user interface (TUI) from the controller single-board computer (SBC) serial console from ten hierarchical and separate-but-related command-line interface (CLI) programs:

- *setup*
  - *setup-station*
  - *setup-system*
  - *setup-network*
    - *setup-ethernet*
    - *setup-wifi*
    - *setup-cellular*
    - *setup-satellite*
    - *setup-preferred-network*
    - *setup-server*

Generally, for first-time controller *out-of-box* installation configuration and setup, as of firmware release `3.0.0` and later, `setup` is the only program installers need to or should interact with.

Subsequent to first-time configuration and setup, installers can choose the setup program that most closely meets their specific tuning or customization needs.

These programs are summarized in Table 1 below.

Setup Command	Description	Hierarchical?	Reboot Required?
`setup`	Top-level, first-time controller configuration and setup. This encompasses and runs all of the setup commands below.	Yes	Yes
`setup-station`	Installation- and site-specific configuration and setup.	No	Yes
`setup-system`	Manifold- and hydraulic-specific configuration and setup.	No	Yes
`setup-network`	Network-specific configuration and setup. This encompasses and runs all of the setup commands below.	Yes	Yes
`setup-ethernet`	Ethernet network-related configuration and setup. Interactively confirms a working Ethernet connection.	No	No
`setup-wifi`	Wi-Fi network-related configuration and setup. Wi-Fi networks are scanned continuously in the background and are presented interactively to choose among, largely obviating the need to manually enter in Wi-Fi network name (that is, SSID). Interactively confirms a working Wi-Fi connection.	No	No
`setup-cellular`	Cellular network-related configuration and setup. Interactively confirms a working Cellular connection.	No	No
`setup-satellite`	Satellite network-related configuration and setup.	No	Yes
`setup-preferred-network`	Preferred network configuration and setup. The controller preferred network	No	Yes

	is that network technology which, if not serving as the primary network interface, will generate an alert to the Frontline service.		
`setup-server`	Optional Frontline cloud services configuration and setup. This may be invoked directly or using the `--with-server` option to `setup` or `setup-network`. This should **only** be used if **explicitly requested** by product engineering or support.	No	Yes

Table 1. Summary of available controller setup commands.

setup

Generally, for first-time controller *out-of-box* installation configuration and setup, `setup` is the only program installers need to or should interact with.

`setup` is hierarchical and includes the following three setup commands:

- *setup-station*
- *setup-system*
- *setup-network*

eliminating the need to run them individually. The details of each are explained below.

setup-station

This setup program focuses on premises site installation information and a subset of network configuration, including:

- Customer Name
- Install Location
- Install Date

If it is not run as part of `setup`, `setup-station` must be run **first**, before `setup-system`, and, at present, requires rebooting the system upon completion with the `reboot` CLI command.

After initial, first-time setup, `setup-station` may be run at any time subsequently to change any of the settings it covers.

`setup-station` **is** *non-destructive* in that it **will** read and preserve existing settings, if they exist, and offer them up as default input.

setup-system

This setup program focuses on hydraulic hardware configuration (HHC), and activation and hydration, including:

- Hydraulic Hardware Configuration
  - Flow Sensor
  - Foam Module
  - System Valves
  - Zone Setup
- Activation and Hydration
  - Drain Cycle
  - Initiation Cycles
  - Water Use Mode

If it is not run as part of `setup`, `setup-system` must be run **second**, after `setup-station`, and, at present, requires rebooting the system upon completion with the `reboot` CLI command. After initial, first-time setup, `setup-system` may be run at any time subsequently to change any of the settings it covers.

Unlike `setup-station`, `setup-system` **is** *destructive* in that it **will not** read and preserve existing settings, if they exist. Consequently, care should be taken to note existing settings before running `setup-system` after initial setup.

setup-network

This setup program focuses on premises site installation information and a subset of network configuration, including:

- Ethernet
- Wi-Fi
- Cellular
- Satellite
- Primary Internet Network Interface Type
- Frontline Cloud Services

If it is not run as part of `setup`, `setup-network` **must** be run **third**, after `setup-system`, and, at present, requires rebooting the system upon completion with the `reboot` CLI command. After initial, first-time setup, `setup-network` may be run at any time subsequently to change any of the settings it covers.

Unlike in the past with the controller, beyond the *Primary Internet Network Interface Type* Frontline Cloud Services alert designation, there is no longer any distinction of backup interfaces. Any configured network technology can be used at any time, as long as it has connectivity. Generally, if available, configured, and connected, the controller will prefer network technologies in this order:

1. Ethernet

2. Wi-Fi
3. Cellular
4. Satellite

`setup-network` is hierarchical and includes the following six setup commands:

- *setup-ethernet*
- *setup-wifi*
- *setup-cellular*
- *setup-satellite*
- *setup-preferred-network*
- *setup-server*

eliminating the need to run them individually. The details of each are explained below.

setup-ethernet

This setup program focuses exclusively on Ethernet configuration.

`setup-ethernet` interactively confirms a working Ethernet connection. So, as an installer, you know whether the Ethernet connection is usable before the command terminates.

After using `setup-ethernet`, `ethernet-check` may always be run to interactively check the Ethernet connection.

`setup-ethernet` may be run at any time, in any order, and does **not** require the `reboot` CLI command for the system to pick up changes made with the program. However, if the primary Internet network interface is changed as a result of using `setup-ethernet`, `setup-preferred-network` must be run to confirm that the correct primary Internet network interface is still set. Failure to do so, may result in unintended alerts to the Frontline Cloud Services regarding controller connectivity.

setup-wifi

This setup program focuses exclusively on Wi-Fi configuration.

`setup-wifi` continuously and periodically scans Wi-Fi networks in the background. The networks it finds are presented interactively to choose among, largely eliminating the need to manually enter in Wi-Fi network name (that is, SSID) as has been the case with the controller in the past.

In Wi-Fi passwords (that is, pre-shared keys (PSKs)) are interactively checked for correctness and validity with interactive feedback provided when incorrect or invalid. Consequently, when and if Wi-Fi password entry fails, you will know whether it was too short, too long, or invalid (that is, not accepted by the Wi-Fi network).

When interacting with the dynamically-generated list of Wi-Fi networks, there are some additional annotations beyond the Wi-Fi network names to note that will help choose among the available networks:

- `(C)` indicates a Wi-Fi network the controller is connected to currently.
- `(F)` indicates a *favorite* Wi-Fi network the controller has connected to previously.
- `(S)` indicates a *secure* (that is, non-open) Wi-Fi network.
- `(=====)` indicates the relative Wi-Fi network signal strength.
    - `(     )` indicates 0% or no Wi-Fi network signal strength.
    - `(=    )` indicates 1-20% Wi-Fi network signal strength.
    - `(==   )` indicates 21-40% Wi-Fi network signal strength.
    - `(===  )` indicates 41-60% Wi-Fi network signal strength.
    - `(==== )` indicates 61-80% Wi-Fi network signal strength.
    - `(=====)` indicates 81-100% Wi-Fi network signal strength.

Generally, Wi-Fi networks with two “bars” or fewer **should not** be used for controller Wi-Fi connections. Wi-Fi networks with “three” bars may have issues. For such networks, use `wifi-check` after setup to see a more exact signal strength and how the Wi-Fi network performs.

Networks are sorted by:

1. Any that the controller is connected to now (that is, those noted with `(C)`).
2. Any that the controller has connected to previously (that is, those noted with `(F)`).
3. Signal strength.
4. Network name.

`setup-wifi` interactively confirms a working Wi-Fi connection. So, as an installer, you know whether the Wi-Fi connection is usable before the command terminates.

As noted above, after using `setup-wifi`, `wifi-check` may always be run to interactively check the Wi-Fi connection.

`setup-wifi` may be run at any time, in any order, and does **not** require the `reboot` CLI command for the system to pick up changes made with the program. However, if the primary Internet network interface is changed as a result of using `setup-wifi`, `setup-preferred-network` must be run to confirm that the correct primary Internet network interface is still set. Failure to do so, may result in unintended alerts to the Frontline Cloud Services regarding controller connectivity.

setup-cellular

This setup program focuses exclusively on Cellular configuration.

`setup-cellular` interactively confirms a working Cellular connection. So, as an installer, you know whether the Cellular connection is usable before the command terminates.

After using `setup-cellular`, `cellular-check` may always be run to interactively check the Cellular connection.

`setup-cellular` may be run at any time, in any order, and does **not** require the `reboot` CLI command for the system to pick up changes made with the program. However, if the primary Internet network interface is changed as a result of using `setup-cellular`, `setup-preferred-network` must be run to confirm that the correct primary Internet network interface is still set. Failure to do so, may result in unintended alerts to the Frontline Cloud Services regarding controller connectivity.

setup-satellite

This setup program focuses exclusively on the Iridium Short Burst Data (SBD) Satellite configuration.

Unlike the other commands, `setup-satellite` does **not** interactively confirm a working Satellite connection. Consequently, after using `setup-satellite`, `satellite-check` **should always** be run to interactively check the Satellite connection.

If it is not run as part of `setup` or `setup-network` , `setup-satellite` **must** be run after `setup-system`, and, at present, **requires** rebooting the system upon completion with the `reboot` CLI command.

After initial, first-time setup, `setup-satellite` may be run at any time subsequently to change any of the settings it covers.

setup-preferred-network

This setup program focuses exclusively on:

- Primary Internet Network Interface Type

The *Primary Internet Network Interface Type* is that controller network technology which, if not serving as the primary network interface, will generate an alert to the Frontline service. So, for example, if you choose Ethernet as *Primary Internet Network Interface Type* but it is disabled or not connected and the controller primarily is on Wi-Fi, then an alert will be generated for the controller with the Frontline service until *Primary Internet Network Interface Type* is changed or until Ethernet is re-enabled and re-connected.

`setup-preferred-network` tries to make this setup process easy by offering the most appropriate network technology as the default choice based on its availability, configuration, and connectivity. Generally, unless there is a compelling reason to do otherwise, just hit the `Enter` or `Return` key and accept the default choice suggested by `setup-preferred-network`.

setup-server

This covers configuration and setup of *optional* Frontline cloud services that are normally handled **automatically** in the background.

If you are explicitly requested to do so by customer / field support or product engineering, this may be invoked directly or using the `-with-server` option to `setup` or `setup-network`. If so requested, customer / field support or product engineering will instruct what to enter. If you inadvertently or accidentally invoke this command, you can just accept the default. Should you attempt to enter anything other than the default, `setup-server` will *speed bump* you and confirm that you really want to use what was entered.

Process

Equipment and Tools

To run setup on a controller, you will need the following equipment or tools:

1. Required
    a. A laptop or tablet device with a USB Type A or USB Type C port.
        i. Frontline recommends the Microsoft Surface.
    b. USB Type A or C to DB9 Female serial adapter cable.
        i. Frontline recommends the DTech 10' USB 2.0 to RS232 DB9 Serial Port Adapter Cable with FTDI Chipset from Amazon.
    c. A terminal app or program for your laptop or tablet device.
        i. Frontline recommends PuTTY from the Microsoft App Store.
    d. #1 Phillips head screwdriver.

Steps

Setup

To run setup on a controller, perform the following top-level steps:

1. With a #1 Phillips head screwdriver, remove the lower, motor control board (MCB) protective acrylic panel by removing the four screws securing it to their mounting studs and then remove the panel.
2. Connect the USB Type A or C to DB9 Female serial adapter cable to the serial console DB9 male port on the MCB, labeled *Console* on the MCB silk screen, as shown in Figure 1 below.
3. Launch the terminal app or program on your laptop or tablet device.
    a. While using the particular terminal app or program on your laptop or tablet device is beyond the scope of this document, the controller serial port requires a serial port speed of 115,200 bits per second (bps).
4. Power on the controller and/or SBC by connecting the power source.
5. At the `fwds login:` log in prompt, enter the user name `root`, as shown in Listing 1 below.
6. At the `Password:` prompt, enter the `root` password.
7. At the `<year>-<month>-<day>T<hour>:<minute>:<second><timezone> [<serial number]#` prompt, enter the `setup` command followed by `return` or `enter` key on your tablet or laptop, as shown in Listing 2 below.

8. Follow the prompts of `setup`.
9. Reboot the controller by entering the `reboot` command followed by *return* or *enter* key on your tablet or laptop, as shown in Listing 3 below.
10. Once the controller has rebooted and returned to the `fwds login:` log in prompt, repeat steps 5 and 6 above.
11. Confirm that the controller has successfully started up by confirming a steady, green on the controller front panel "*READY*" status LED[1].
12. Confirm that the controller has successfully connected to the Frontline cloud service over one of Ethernet, Wi-Fi, or Cellular by confirming a steady, green on the controller front panel "*ONLINE*" status LED.
13. Remove the USB Type A or C to DB9 Female serial adapter cable from the serial console DB9 male port on the MCB.
14. With a #1 Phillips head screwdriver, re-install the lower, motor control board (MCB) protective acrylic panel.

Figure 1. Controller serial console in the upper left area of the motor control board (MCB).

```
Frontline Wildfire Defense Systems (FWDS) Controller - fwds /dev/ttymxc3

fwds login: root
```

Listing 1. Controller serial console log in prompt.

```
2024-01-01T12:00:00-0600 [36002422]# setup
```

[1] See the "*Frontline Controller LEDs*" in *References* below for more information about controller front panel LEDs.

Listing 2. Initiating `setup` from the controller root prompt.

```
2024-01-01T12:00:30-0600 [36002422]# reboot
```

Listing 3. Initiating `reboot` from the controller root prompt.

Setup

Station Setup

As briefly discussed above, this setup program focuses on premises site installation information and a subset of network configuration, including:

- Customer Name
- Install Location
- Install Date

Generally, you should have discussed the inputs and answers to these `setup` topics and prompts with the premises owner or site manager and/or Frontline project manager prior to performing setup.

The general format for the prompts in `setup` is shown in Listing 4 below and includes:

- The prompt topic or question.
- The prompt topic or question options or format, in parenthesis.
- The current or default setting or value, in brackets.
    - This is the setting or value that will be accepted if you simply hit the *return* or *enter* key on your tablet or laptop.
- A colon (':') or question mark ('?').
- The current or default setting or value, which is editable using backspace or arrow keys.
    - To accept it as-is, simply hit the `return` or `enter` key on your tablet or laptop.

```
Topic / Question (Format / Options) [Current / Default Value]: Editable Current /
Default Value
```

Listing 4. Format of `setup-station` prompts topics or questions.

To run through the prompts for `setup` on a controller, do the following:

1. Enter the ***Customer name***:
    a. From the factory, this defaults to "*Not Yet Deployed*".
    b. Enter the premises owner or manager first and last name.
2. Enter the ***Location***:
    a. From the factory, this defaults to "*NA*".

   b. If there is only one controller on the premises, enter the premises address, separated by semicolons ';'. For example, *10555 Quality Road; Quality, CA 93215*.
   c. If there are multiple controllers on the premises, enter a descriptive location for the controller. For example, "*Main House*" or "*Garage*".
3. Enter the ***Installation date***:
   a. This defaults to the current date in year-month-day format.

```
2024-01-01T12:00:00-0600 [36002422]# setup
Running setup...
    Running station setup...
        Customer name [Not Yet Deployed]: Will Durness
        Location [NA]: 10555 Quality Road; Quality, CA 93215
        Installation date (YYYY-MM-DD) [2024-01-01]: 2023-11-01
...
```

Listing 5. Running through the *station*-related `setup` prompts.

System Setup

After the *station* phase of `setup` is complete, `setup` will transition into *system* setup. As briefly discussed above, this portion of setup focuses on a subset of hydraulic hardware configuration (HHC), and activation and hydration, including:

- Hydraulic Hardware Configuration
  - Flow Sensor
  - Foam Module
  - System Valves
  - Zone Setup
- Activation and Hydration
  - Drain Cycle
  - Initiation Cycles
  - Water Use Mode

Generally, you should be familiar with, through the process of onsite installation, and/or have discussed the inputs and answers to these `setup` topics and prompts with the premises owner or site manager and/or Frontline project manager prior to performing setup.

The general format for the prompts in this phase of `setup` is the same as for the prior phase, as shown in Listing 5 above. All phases of `setup` uses indentation to group related topics or questions together as shown in the listings below.

To run through the prompts for this phase of `setup` on a controller, do the following:

1. Choose the **Manifold hydraulic hardware configuration (HHC)**, as shown in Listings 6-8 below:
    a. The default is *MP3*.
    b. This has a direct impact on how the controller manages the system and zone valves and sensors; consequently, answering this correctly is **critically important** to correct system operation.
    c. The answer to this prompt will determine the remainder of the prompts.
    d. When and wherever possible, setup-system attempts to intuit the answers to many setup topics and will indicate it has done so such as *`Running flow sensor setup for HHC "mp3"....`*
2. Regardless of the HHC selected, as shown in Listings 6-8 below:
    a. Choose the **Number of as-built zones**.
        i. While the controller supports up to nine (9) zones, the controller should be configured only for the number of zones actually built and installed.
        ii. For each zone:
            1. Choose the **Zone type**, either **`Roof`**, **`Eave`**, or **`Perimeter`**.
                a. This has a direct impact on how the controller operates the zone during activation and hydration; consequently, answering this correctly is **critically important** to correct system operation.
            2. Choose the **Zone name**:
                a. This defaults to the zone type and number, for example "*Roof Zone 1*".
                b. This should be descriptive of where the zone is located such that during activation or service, the customer, property manager, and/or Frontline field and support personnel can correlate the zone in the field or in the Frontline app with where it is on the premises.
                c. This is limited to 16 characters.
    b. Choose the **Number of initiation cycles**:
        i. The default is four (4).
        ii. This is the number of cycles (or times) that the controller will run through a hydration profile *Initiation* cycle type when activated before transitioning to *Maintenance* cycle types. Initiation cycles put down more water and foam than Maintenance cycles to ensure the premises is well hydrated before maintaining that hydration.
        iii. The Frontline project designer and manager will have made the appropriate determination and designation for site-specific considerations as to this setting.
        iv. If you are unsure, accept the default value.
    c. Choose the **Water use mode**:

        i. The default is `Standard`.
        ii. The Frontline project designer and manager will have made the appropriate determination and designation for site-specific considerations as to this setting.
        iii. If you are unsure, accept the default value.
3. If *MP3* was chosen as the HHC, as shown in Listing 6:
    a. Choose whether the manifold has an installed foam system module.
        i. The default is `Yes`.
    b. Choose whether to Run a drain cycle during deactivation.
        i. The default is `Yes`.
        ii. MP3 HHCs have a Drain valve that is capable, in conjunction with zone valves, of draining the hydraulic system of fluid during deactivation. This process takes about two (2) minutes and happens, if `Yes` is chosen, at the end of the deactivation process.
        iii. If you are unsure, accept the default value.
4. If *HP6* was chosen as the HHC, as shown in Listing 7:
    a. Choose whether to **Use the bypass valve in addition to the foam valve when running a foam profile**:
        i. The default is `No`.
        ii. In HP6 (and Legacy) HHCs, the controller is capable of opening both the Bypass (or Water) and Foam valves during activation for foam-based profiles to generate extra water flow.
        iii. The Frontline project designer and manager will have made the appropriate determination and designation for site-specific considerations as to this setting.
        iv. If you are unsure, accept the default value.
    b. Choose whether to **Enable pressure relief support**:
        i. The default is `Yes`.
        ii. In certain circumstances for HP6 (and Legacy) HHCs, pressure that is harmful to the hydraulic system may build up in the controller during activation. The controller is able to open a designated zone valve to relieve this pressure.
        iii. If `Yes` is selected, choose the **Zone number to use for pressure relief**:
            1. The default is one (1).
        iv. The Frontline project designer and manager will have made the appropriate determination and designation for site-specific considerations as to these settings.
        v. If you are unsure, accept the default values.
5. If `Legacy` was chosen as the HHC, as shown in Listing 8:
    a. Choose the **system valve type** for each of the *Master*, *Bypass*, and *Foam* valves:
        i. This has a direct impact on how the controller manages the system valves; consequently, answering this correctly is **critically important** to correct system operation.

b. Choose whether to **Use the bypass valve in addition to the foam valve when running a foam profile**:
   i. The default is `No`.
   ii. In Legacy (and HP6) HHCs, the controller is capable of opening both the Bypass (or Water) and Foam valves during activation for foam-based profiles to generate extra water flow.
   iii. The Frontline project designer and manager will have made the appropriate determination and designation for site-specific considerations as to this setting.
   iv. If you are unsure, accept the default value.
c. For each zone, choose the **zone valve type**:
   i. This has a direct impact on how the controller manages the zone valves; consequently, answering this correctly is **critically important** to correct system operation.
d. Choose whether to **Enable pressure relief support**:
   i. The default is `Yes`.
   ii. In certain circumstances for HP6 (and Legacy) HHCs, pressure that is harmful to the hydraulic system may build up in the controller during activation. The controller is able to open a designated zone valve to relieve this pressure.
   iii. If `Yes` is selected, choose the **Zone number to use for pressure relief**:
      1. The default is one (1).
   iv. The Frontline project designer and manager will have made the appropriate determination and designation for site-specific considerations as to these settings.
   v. If you are unsure, accept the default values.

```
...
    Running system setup...
        Manifold hydraulic hardware configuration (HHC) (0-Legacy, 1-HP6, 2-MP3, 3-LV2) [2]: 2
        Running manifold setup for HHC "mp3"...
            Running foam module setup for HHC "mp3"...
                Does the HHC manifold have an installed foam system module (0|N-No, 1|Y-Yes) [Y]? Y
            Running valve setup for HHC "mp3"...
                Running system valve setup for HHC "mp3"...
                    Running "Drain" valve setup for HHC "mp3"...
                    Running "Water" valve setup for HHC "mp3"...
                    Running "Foam" valve setup for HHC "mp3"...
                Running zone valve setup for HHC "mp3"...
                    Number of as-built zones (1-9): 3
                    Running zone 1 setup for HHC "mp3"...
                        Zone 1 type (0-Roof, 1-Eave, 2-Perimeter): 0
                        Zone 1 name [Roof Zone 1]: Garage Roof
```

```
                            Running "Roof Zone 1" valve setup for HHC "mp3"...
                    Running zone 2 setup for HHC "mp3"...
                        Zone 2 type (0-Roof, 1-Eave, 2-Perimeter): 1
                        Zone 2 name [Eave Zone 2]: Second Story Eaves
                            Running "Eave Zone 2" valve setup for HHC "mp3"...
                    Running zone 3 setup for HHC "mp3"...
                        Zone 3 type (0-Roof, 1-Eave, 2-Perimeter): 2
                        Zone 3 name [Eave Zone 3]: Fence Line
                            Running "Eave Zone 3" valve setup for HHC "mp3"...
            Run a drain cycle during deactivation (0|N-No, 1|Y-Yes) [Y]? Y
        Number of initiation cycles [4]: 4
        Water use mode (0-Standard, 1-High) [0]: 0
...
```

Listing 6. Running through the *system*-related `setup` prompts for *MP3* HHCs.

```
...
    Running system setup...
        Manifold hydraulic hardware configuration (HHC) (0-Legacy, 1-HP6, 2-MP3, 3-LV2) [2]: 1
        Running manifold setup for HHC "hp6"...
            Running valve setup for HHC "hp6"...
                Running system valve setup for HHC "hp6"...
                        Running "Master" valve setup for HHC "hp6"...
                        Running "Bypass" valve setup for HHC "hp6"...
                        Running "Foam" valve setup for HHC "hp6"...
                    Use the bypass valve in addition to the foam valve when running a foam profile (0|N-No,
1|Y-Yes) [N]? N
                Running zone valve setup for HHC "hp6"...
                    Number of as-built zones (1-9): 3
                    Running zone 1 setup for HHC "hp6"...
                        Zone 1 type (0-Roof, 1-Eave, 2-Perimeter): 0
                        Zone 1 name [Roof Zone 1]: Second Story Roof
                            Running "Second Story Roof" valve setup for HHC "hp6"...
                    Running zone 2 setup for HHC "hp6"...
                        Zone 2 type (0-Roof, 1-Eave, 2-Perimeter): 0
                        Zone 2 name [Roof Zone 2]: First Story Roof
                            Running "First Story Roof" valve setup for HHC "hp6"...
                    Running zone 3 setup for HHC "hp6"...
                        Zone 3 type (0-Roof, 1-Eave, 2-Perimeter): 1
```

```
                        Zone 3 name [Eave Zone 3]: Patio Eaves
                            Running "Patio Eaves" valve setup for HHC "hp6"...
            Enable pressure relief support (0|N-No, 1|Y-Yes) [Y]? Y
                Zone number to use for pressure relief (1-3): 1
        Number of initiation cycles [4]: 4
        Water use mode (0-Standard, 1-High) [0]: 0
...
```

Listing 7. Running through the *system*-related `setup` prompts for *HP6* HHCs.

```
...
    Running system setup...
        Manifold hydraulic hardware configuration (HHC) (0-Legacy, 1-HP6, 2-MP3, 3-LV2) [2]: 0
        Running manifold setup for HHC "legacy"...
            Running valve setup for HHC "legacy"...
                Running system valve setup for HHC "legacy"...
                        Running "Master" valve setup for HHC "legacy"...
                            "Master" valve type (B-Ball, S-Solenoid): S
                        Running "Bypass" valve setup for HHC "legacy"...
                            "Bypass" valve type (B-Ball, S-Solenoid) [S]: S
                        Running "Foam" valve setup for HHC "legacy"...
                            "Foam" valve type (B-Ball, S-Solenoid) [S]: S
                    Use the bypass valve in addition to the foam valve when running a foam profile (0|N-No,
1|Y-Yes) [N]? N
                Running zone valve setup for HHC "legacy"...
                    Number of as-built zones (1-9): 1
                    Running zone 1 setup for HHC "legacy"...
                        Zone 1 type (0-Roof, 1-Eave, 2-Perimeter): 0
                        Zone 1 name [Roof Zone 1]: Main Roof
                            Running "Main Roof" valve setup for HHC "legacy"...
                                "Main Roof" valve type (B-Ball, S-Solenoid) [S]: S
            Enable pressure relief support (0|N-No, 1|Y-Yes) [Y]? Y
                Zone number to use for pressure relief (1-1): 1
        Number of initiation cycles [4]: 4
        Water use mode (0-Standard, 1-High) [0]: 0
...
```

Listing 8. Running through the *system*-related `setup` prompts for *Legacy* HHCs.

As noted above, any changes made via the *system* phase of `setup` require a reboot of the controller.

Network Setup

After the *system* phase of `setup` is complete, `setup` will transition into network setup. As briefly discussed above, this portion of setup focuses on network configuration and setup, including:

- Ethernet
- Wi-Fi
- Cellular
- Satellite
- Primary Internet Network Interface Type

As shown below in Listing 9, network setup for the controller is designed to be as easy and straightforward as possible. If you have set up network settings on your mobile phone, tablet, or computer, the network setup on the controller should feel familiar.

As noted above, when prompted for the *Primary Internet Network Interface Type*, accepting the default prompt choice is recommended.

```
...
        Running network setup...
        Setting up Ethernet...
            Checking Ethernet availability...
                Ethernet is available.
                Enable Ethernet networking (0|N-No, 1|Y-Yes) [Y]? Y
                    Configuring IP addresses...
                    Confirming Internet reachability...
                Successfully connected to Ethernet network.
        Setting up Wi-Fi...
            Checking Wi-Fi availability...
                Wi-Fi is available.
                Enable Wi-Fi networking (0|N-No, 1|Y-Yes) [Y]? Y

                        [R]  Refresh and re-display Wi-Fi networks

                        Choose a Wi-Fi network option from the above (R) [R]: R

                        [ 1]            (S)  (=====)  Southern Cousin
```

```
                    [ 2]            (S)  (=====)  Robust Debt
                    [ 3]            (S)  (==== )  Deranged Love
                    [ 4]            (S)  (==== )  Rare Goal
                    [ 5]            (S)  (===  )  Energetic Steak
                    [ 6]            (S)  (===  )  Miniature Moment
                    [ 7]            (S)  (==   )  Delightful Owner
                    [ 8]                 (==   )  Superficial Role
                    [ 9]            (S)  (=    )  Chemical Message
                    [10]            (S)  (=    )  Crowded Language

                    [R]  Refresh and re-display Wi-Fi networks

                    Choose a Wi-Fi network option from the above (1-5, R) [R]: 1
                        Connecting to the Wi-Fi network "Southern Cousin"...
                        Enter the passphrase for the Wi-Fi network "Southern Cousin": 6ae7-ecd9-494a
                        Connecting to the Wi-Fi network "Southern Cousin"...
                            Associating with the Wi-Fi network "Southern Cousin" access point...
                            Configuring IP addresses...
                        Successfully connected to the Wi-Fi network "Southern Cousin".
Setting up Cellular...
    Checking Cellular availability...
        Cellular is available.
        Enable Cellular networking (0|N-No, 1|Y-Yes) [Y]? Y
            Configuring IP addresses...
        Successfully connected to Cellular network.
Setting up Satellite...
    Checking Satellite availability...
        Satellite is available.
        Enable Satellite networking (0|N-No, 1|Y-Yes) [Y]? Y
Running preferred network setup...
    Preferred network type (E-Ethernet, W-Wi-Fi, C-Cellular) [E]: E
```

Listing 9. Running through the *network*-related `setup` prompts.

As noted above, some changes made via the network phase of `setup` require a reboot of the controller.

References

1. Frontline Wildfire Defense Systems, Inc. "*Controller Siting & Install Manual*". 2023-01-17. Document.
2. Frontline Wildfire Defense Systems, Inc. "*Frontline Controller LEDs*". Revision 4. 2023-01-12. Document.
3. Frontline Wildfire Defense Systems, Inc. "*Terminology Glossary*". Revision 2. 2022-06-29.

Revision History

Revision	Date	Modified By	Description
1	2023-11-30	Grant Erickson	Initial revision.
2	2023-12-21	Grant Erickson	Fixed minor typos.
3	2024-05-31	Grant Erickson	Updated for LV2 and 3.0.0.
4	2025-10-27	Grant Erickson	Added `Perimeter` zone type.